LAB - 05

SVM

- SVR → for regression
- SVC → for classification

SVM → separate data points.
→ plot two lines.

cv : 5 cross-validation

↳ Divide the data into 5 folds.

eg:. on training apply testing as well.

↖ before we are using training / testing

Now cv= 5

Parameters.

- C ⇒ how much penalty allow? Margin for mistake → 0.01 margin
- Kernel. ⇒ poly kernel → degree. control.
- degree
- gamma.
- According to dataset, set the kernel.

Kernels → RBF → mostly used. → gamma values
→ Sigmoid → curve ∫

Grid Search CV

↳ check all the combinations (using diff kernels)

↳ Then suggest the best one (optimal).

Voting Classifier

- through sklearn import voting Classifier
- sklearn . ensemble.
- use multiple models & provide the best one.
  ↙ voting classifier.

K - Mean Clustering (Classification Model)

- k = 3 → 3 groups.
- centroid → base for one group.
- Distance → Euclidean Distance.
- Minimum distance → centroid. → move to one side & combine.
  centroid
- Takes the average of all the dataset points
  ↳ Then assign new position of centroid